Relations	Template #1	Template #2	Template #3
P176 (`manufacturer`)	[X] is produced by [Y]	[X] is a product of [Y]	[Y] and its product [X]
P463 (`member_of`)	[X] is a member of [Y]	[X] belongs to the organization of [Y]	[X] is affiliated with [Y]
P407 (`language_of_work`)	[X] was written in [Y]	The language of [X] is [Y]	[X] was a [Y]-language work

Table 1: Example prompt templates of three relations in PARAREL. [X] and [Y] are the placeholders for the head and tail entities, respectively. Owing to the page width, we show only three templates for each relation. Prompt templates in PARAREL produce 253,448 knowledge-expressing prompts in total for 27,738 relational facts.

4 Experiments

4.1 Experimental Settings

We conduct experiments for BERT-base-cased (Devlin et al., 2019), one of the most widely-used pretrained models. It contains 12 Transformer blocks, where the hidden size is 768 and the FFN inner hidden size is 3,072. Notice that our method is not limited to BERT and can be easily generalized to other pretrained models. For each prompt, we set the attribution threshold t to 0.2 times the maximum attribution score. For each relation, we initialize the refining threshold $p\%$ (Section 3.3) as 0.7. Then, we increase or decrease it by 0.05 at a time until the average number of knowledge neurons lies in [2, 5]. We run our experiments on NVIDIA Tesla V100 GPUs. On average, it costs 13.3 seconds to identify knowledge neurons for a relational fact with 9 prompts.

4.2 Dataset

We examine knowledge neurons through the fill-in-the-blank cloze task based on the PARAREL dataset (Elazar et al., 2021). PARAREL is curated by experts, containing various prompt templates for 38 relations from the T-REx dataset (ElSahar et al., 2018). We show some example templates in Table 1. For each relational fact, we fill in the head entity in prompt templates and leave the tail entity as a blank to predict. In order to guarantee the template diversity, we filter out relations with fewer than 4 prompt templates and finally keep 34 relations, where each relation has 8.63 different prompt templates on average. These prompt templates produce 253,448 knowledge-expressing prompts in total for 27,738 relational facts.

4.3 Attribution Baseline

Our baseline method takes the neuron activation value as the attribution score, i.e., $\mathrm{Attr}_{\mathrm{base}}(w_i^{(l)}) = \overline{w}_i^{(l)}$, which measures how sensitive a neuron is to the input. After computing attribution scores, we follow the same pipeline to obtain the refined knowledge neurons. For a fair comparison, we employ the same method to choose the hyper-parameters t and $p\%$ for the baseline to ensure the average number of knowledge neurons for each relation lies in $[2, 5]$.

The method based on neuron activation is a reasonable baseline. It is motivated by FFNs's analogy with the self-attention mechanism (as described in Section 2), because self-attention scores are usually used as a strong attribution baseline (Kovaleva et al., 2019; Voita et al., 2019; Hao et al., 2021).

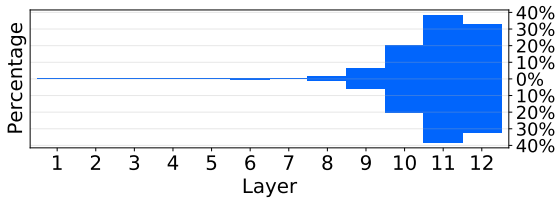

Figure 3: Percentage of knowledge neurons identified by our method in each Transformer layer.

Type of Neurons	Ours	Baseline
Knowledge neurons	4.13	3.96
$\bigcap$ of intra-rel. fact pairs	1.23	2.85
$\bigcap$ of inter-rel. fact pairs	0.09	1.92

Table 2: Statistics of knowledge neurons. $\bigcap$ denotes the intersection of knowledge neurons of fact pairs. "*rel.*" is the shorthand of relation. Our method identifies more exclusive knowledge neurons.

4.4 Statistics of Knowledge Neurons

Figure 3 presents the layer distribution of knowledge neurons identified by our knowledge attribution method. We notice that most fact-related neurons are distributed in the topmost layers of pretrained Transformers. The finding also agrees with Tenney et al. (2019) and Geva et al. (2020).

Table 2 shows statistics of knowledge neurons. On average, we identify 4.13 knowledge neurons for each relational fact using our knowledge attribution method, and 3.96 using the baseline method.

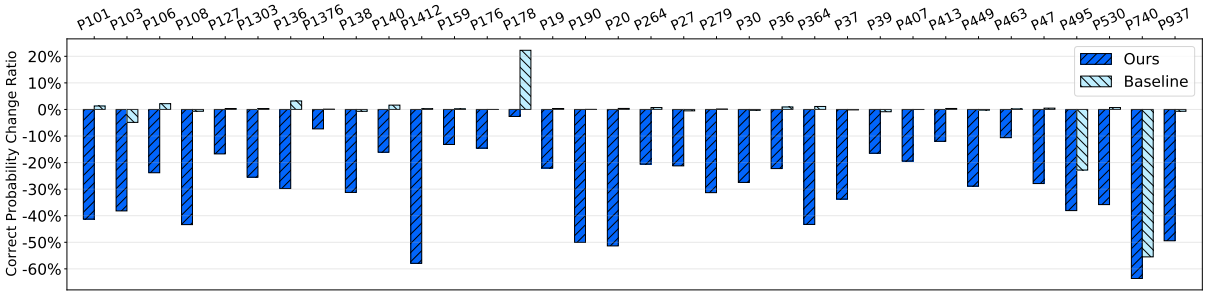

Figure 4: Results of suppressing knowledge neurons for various relations. Suppressing knowledge neurons decreases the correct probability by 29.03% on average. For the baseline, the decreasing ratio is 1.47% on average.

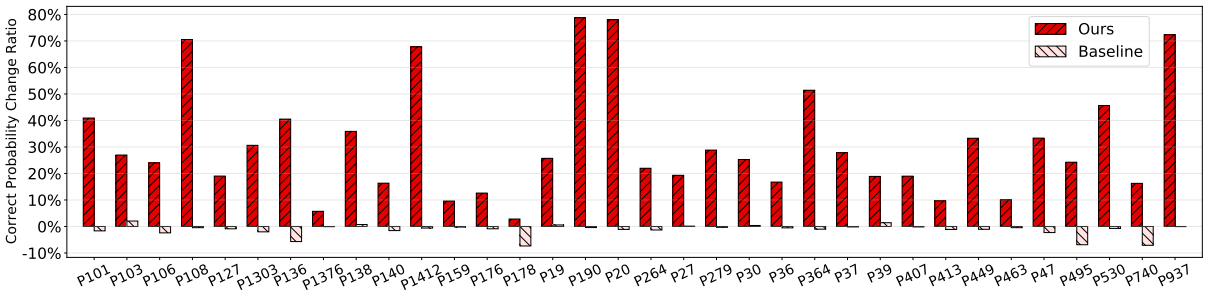

Figure 5: Results of amplifying knowledge neurons for various relations. Amplifying knowledge neurons increases the correct probability by 31.17% on average. For the baseline, the correct probability even decreases by 1.27%.

Their same order of magnitude guarantees the fairness of the subsequent comparisons in the paper.

We also compute the knowledge neuron intersection of different relational facts. Table 2 shows the average number of pair-wise knowledge neuron intersections. For our proposed method, (1) fact pairs with the same relation (*intra-relation fact pairs*) share 1.23 knowledge neurons on average; (2) fact pairs with different relations (*inter-relation fact pairs*) share almost no knowledge neurons. In contrast, for the baseline, (3) most identified neurons are shared by intra-relation fact pairs; (4) even a substantial portion of neurons are common for inter-relation fact pairs. The difference in knowledge neuron intersections suggests that our method can identify more exclusive knowledge neurons.

4.5 Knowledge Neurons Affect Knowledge Expression

We investigate how much knowledge neurons can affect knowledge expression in Figure 4 and Figure 5. Given a relational fact, we manipulate its knowledge neurons in two ways: (1) suppressing knowledge neurons by setting their activations to 0; (2) amplifying knowledge neurons by doubling their activations. Then, for each relation, we plot the average change ratio of the probability for the correct answer, corresponding to the manipulation. For comparison, we also plot the results of manipulating baseline-identified knowledge neurons.

Figure 4 shows that suppressing knowledge neurons identified by our knowledge attribution method leads to a consistent decrease (29.03% on average) in the correct probability. By contrast, for baseline-identified neurons, the suppressing operation has a negligible influence (1.47% decrease on average) on the correct probability. Notably, for the relation P178 (`developer`), the correct probability abnormally increases by using the baseline.

As shown in Figure 5, we have similar observations for amplifying the knowledge neurons identified by our knowledge attribution. We see a consistent increase (31.17% on average) in the correct probability. By contrast, the baseline even decreases the average correct probability by 1.27%.

In summary, the knowledge neurons identified by our knowledge attribution method tend to notably affect knowledge expression. Notice that the above assessment is affected by the distribution of knowledge neurons. For example, if the knowledge neurons for a relation are distributed more widely, we need to manipulate more top-k neurons for better control. We use the above experiments as a proof of concept while leaving precise control for future work.

Relational Facts	Neurons		Top-2 and Bottom-2 Activating Prompts (Average Activation)
⟨ Ireland, capital, Dublin ⟩	$w^{(9)}_{2141}, w^{(10)}_{1122}$	Top	Our trip ... in **Dublin**, the capital and largest city of **Ireland** ... (6.36) **Dublin** is the capital and largest city of **Ireland**. (5.77)
		Bottom	**Dublin** just might be the most iconic destination in all of **Ireland**. (1.27) ... in **Ireland**'s famed city, you can enjoy ... **Dublin** experience ... (-0.30)
⟨ Cao_Yunding, place_of_birth, Shanghai ⟩	$w^{(10)}_{739}, w^{(10)}_{1885}, w^{(11)}_{2876}$	Top	**Cao Yunding** was born in **Shanghai** in November 1989. (3.58) Full name: **Cao Yunding** ... Place of birth: **Shanghai**, China ... (2.73)
		Bottom	... **Cao Yunding** (**Shanghai** Shenhua) is shown the red card ... (-0.30) **Shanghai** Shenhua midfielder **Cao Yunding** ... (-0.31)
⟨ Kuwait, continent, Asia ⟩	$w^{(6)}_{147}, w^{(9)}_{866}, w^{(9)}_{1461}, w^{(10)}_{1169}$	Top	**Kuwait** is thus one of the smallest countries in **Asia** ... (6.63) **Kuwait** is a country in Western **Asia** ... (6.27)
		Bottom	This page displays all **Asia** Society content on **Kuwait** ... (-0.48) Noor **Asia** is ... distribution companies in **Kuwait** ... (-0.59)

Table 3: Example relational facts along with their knowledge neurons, their top-2 and bottom-2 activating prompts, and the corresponding neuron activation. $w^{(l)}_i$ denotes the i-th intermediate neuron at the l-th FFN. We fill the blank in each prompt with the correct answer for better readability. Owing to the page width, we show only key parts for overlong prompts. The top-2 activating prompts express exactly the relation, but the bottom-2 do not.

Prompt Types	Ours	Baseline
Containing head and tail ($\mathcal{T}_1$)	0.485	2.472
Containing only head ($\mathcal{T}_2$)	0.019	2.312
Randomly sampled ($\mathcal{T}_3$)	-0.018	2.244

Table 4: Average activation of knowledge neurons for three types of prompts. The activation of neurons identified by our method can distinguish the knowledge-expressing prompts ($\mathcal{T}_1$) clearly.

4.6 Knowledge Neurons are Activated by Knowledge-Expressing Prompts

In order to study what prompts can activate knowledge neurons, we compare the average activation of knowledge neurons for different types of prompts.

BINGREL Dataset We build a new dataset BINGREL by crawling the Bing search engine to collect new prompts, for a more extensive comparison beyond the PARAREL dataset. For each of the 27,738 facts in PARAREL, we crawl two types of texts: (1) up to ten texts containing both the head and the tail entities (210,217 texts crawled in total); (2) up to ten texts containing only the head entity without restricting tail entities (266,020 texts crawled in total). Following the distant supervision assumption (Mintz et al., 2009), the first type of texts tends to express the whole relational fact, while the second type does not. We mask tail entities for the first type of texts to obtain knowledge-expressing prompts ($\mathcal{T}_1$). In order to conduct a controlled experiment, we mask random words for the second type of texts, forming a control group ($\mathcal{T}_2$). Moreover, we employ randomly sampled prompts as another control group ($\mathcal{T}_3$).

Results As shown in Table 4, for our method, the identified knowledge neurons are more significantly activated by knowledge-expressing prompts ($\mathcal{T}_1 = 0.485$), compared with the control groups ($\mathcal{T}_2 = 0.019$ and $\mathcal{T}_3 = -0.018$). By contrast, for the baseline, the activation of identified neurons cannot distinguish three types of prompts. In addition, since our comparison is based on the web-crawled BINGREL dataset, we validate the generalization of knowledge neurons to open-domain texts that are unseen in PARAREL.

Example Prompts In Table 3, we present example prompts that activate knowledge neurons the most and the least, respectively. Given a fact, we first identify its knowledge neurons with our knowledge attribution method. Then, we calculate the average activation of knowledge neurons for each crawled prompt that contains both the head and the tail entities in BINGREL. Finally, we demonstrate two prompts with the highest average activation values and two with the lowest (denoted as top-2 and bottom-2 activating prompts, respectively).

As shown in Table 3, the top-2 activating prompts express exactly the corresponding relational fact. In contrast, despite containing the same head and tail entities, the bottom-2 activating prompts do not express the correct relation. For example, although the bottom-2 activating prompts for ⟨Ireland, capital, Dublin⟩ express